

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

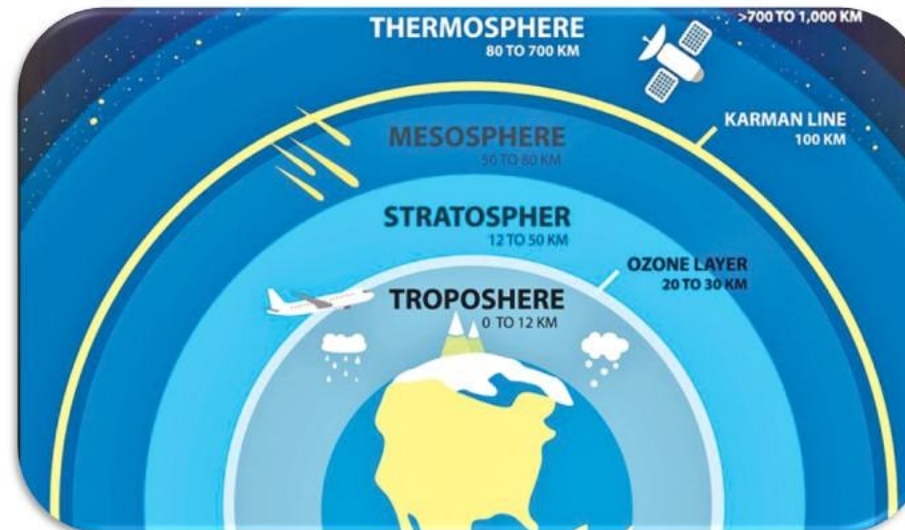
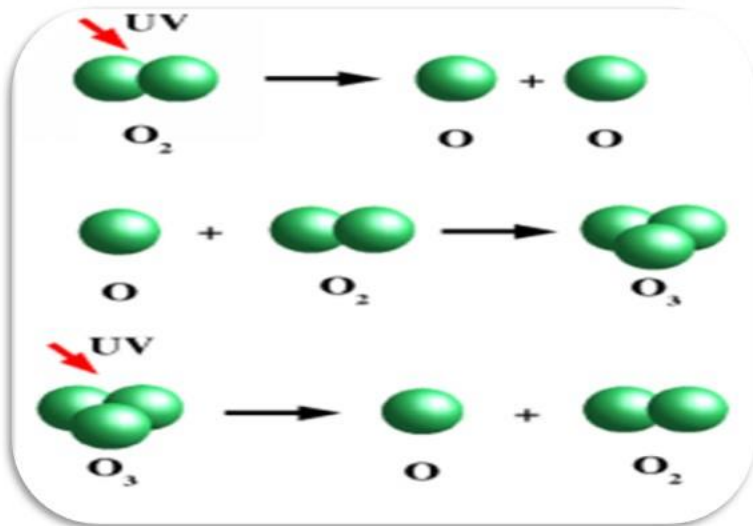
MODULE - 38

OZONE DEPLETION -1

OZONE DEPLETION

THE OZONE LAYER

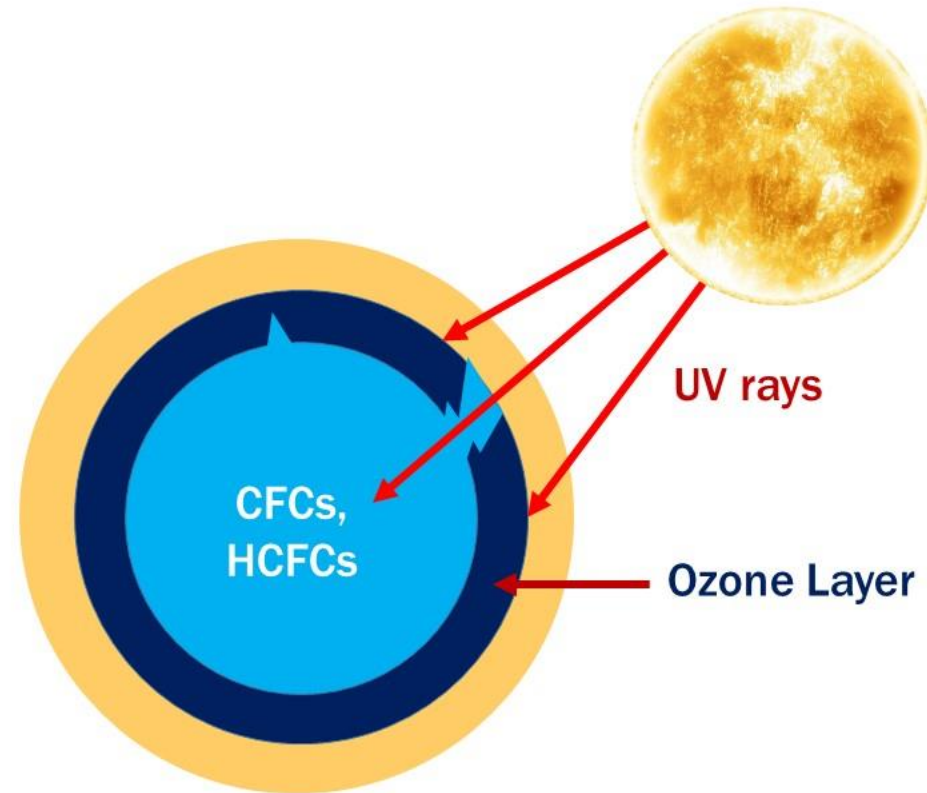
- Ozone Layer - The Ozone Layer is a region in the stratosphere that has large concentrations of ozone (O_3).
- Depending on where ozone is present, it can protect or harm life on Earth. Most ozone is present in the stratosphere where it acts as a shield to protect Earth's surface from the sun's harmful ultraviolet radiation.
- Closer to Earth in the troposphere, ozone is a harmful pollutant that causes damage to lung tissue and plants.



OZONE DEPLETION

THE OZONE LAYER

- Ozone formation and depletion happens naturally in the stratosphere due to the action of UV radiation. Photo-association of oxygen atoms to form ozone and photo-dissociation of ozone into oxygen atoms are continuous processes in the stratosphere.
- When high-energy ultraviolet rays strike ordinary oxygen molecules (O_2), they split the molecule into two single oxygen atoms, known as atomic oxygen.
- Freed oxygen atoms then combine to form a molecule of ozone. Ozone molecule is also split into oxygen atoms by UV radiation.



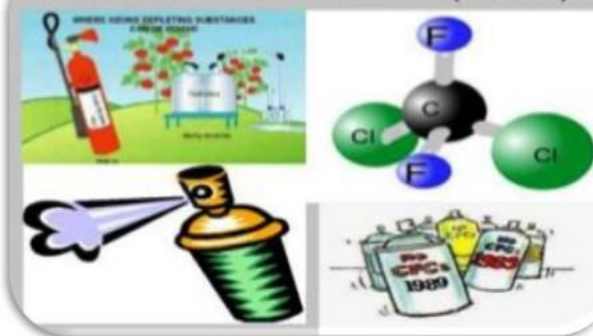
OZONE DEPLETION

OZONE DEPLETING SUBSTANCES

- **Halocarbons** - Halocarbons are compounds in which the hydrogen of a hydrocarbon is replaced by halogens like chlorine, bromine, iodine etc.
- Halogens are a group of reactive non-metallic elements like fluorine, chlorine, bromine, iodine and astatine.
- Halocarbons are gases consisting of carbon and at least one of the halogens (fluorine, chlorine, iodine and bromine).
✓ Example: Chlorofluorocarbons (CFCs).



Chlorofluorocarbons (CFCs)



OZONE DEPLETION

SOME OZONE DEPLETING SUBSTANCES ARE

1. Halocarbon Refrigerants – CFCs and Hydrochlorofluorocarbons (HCFCs)

- CFCs have wide applications because they have a low toxicity, reactivity and flammability.
- CFCs were used as refrigerants for air conditioners and refrigerators.
- These compounds were hailed as a major breakthrough in safety because the first refrigeration systems used ammonia or sulfur dioxide as refrigerants. Ammonia and sulfur dioxide are toxic and corrosive.
- CFCs were also used as the propellants in aerosol spray cans.

2. Halocarbon Solvents – Methyl Chloroform, Carbon Tetra Chloride and Tri-Chloro Ethane.

3. Halons like HydroBromoFluoroCarbons (HBFCs) and BromoChloroMethane previously used in firefighting.

4. Methyl Bromide - a widely used pesticide.

5. Nitrous Oxide (N₂O) also known as 'laughing gas' causes ozone depletion.

MAIN OZONE DEPLETING SUBSTANCES

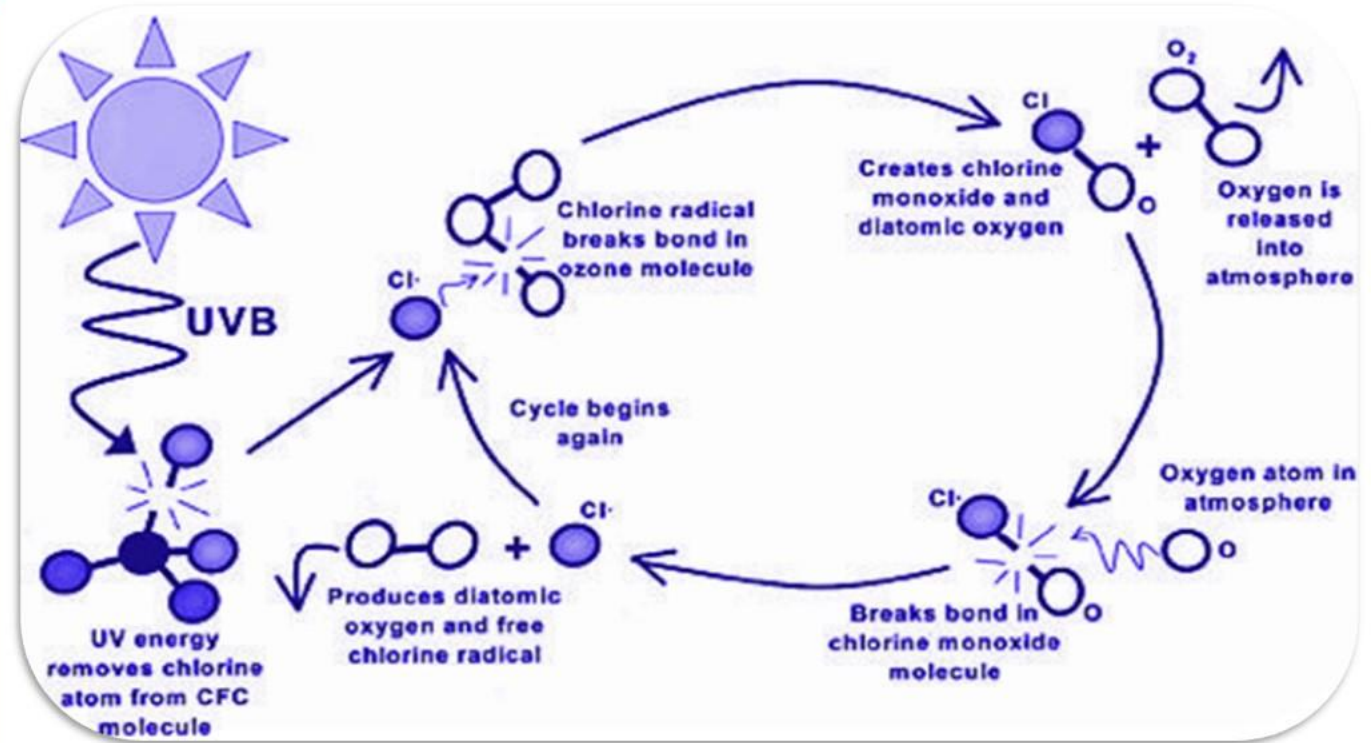


OZONE DEPLETION

OZONE DEPLETING PROCESS

- Halocarbon compounds undergo **Photo-dissociation** (under the influence of UV radiation) which cause them to split up and **release a free halogen radical** (reactive ions).
- These halogen radicals travel up to the stratosphere, **react with ozone layer** and cause ozone depletion

Bromine containing halocarbons are very powerful ozone depleting substances. Each bromine atom destroys 100 times more ozone molecules than what a chlorine atom does.



OZONE DEPLETION

EFFECTS OF OZONE DEPLETION

EFFECTS ON ORGANISMS

- UV rays are highly injurious to living organisms since DNA and proteins of living organisms absorb UV rays and its high energy breaks the chemical bonds within these molecules.
- In humans it causes aging of skin, damage to skin cells and various types of skin cancers.
- In human eye, cornea absorbs UV-B radiation and a high dose of UV-B radiation causes inflammation of cornea and causes snow-blindness, cataract etc.
- UV-B radiation adversely affects the phytoplankton, which in turn adversely affects the marine food chain.
- Coral polyps are very sensitive to UV rays.

EFFECTS OF OZONE DEPLETION

SKIN AILMENTS



DISTURBED PLANT LIFE-CYCLES



DESTRUCTION OF MARINE LIFE



Exposure to UV rays causes coral bleaching.

OZONE DEPLETION

EFFECTS OF OZONE DEPLETION

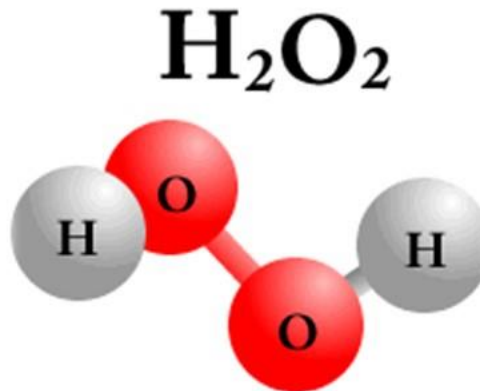
EFFECTS ON PLANTS

- Inhibits photosynthesis
- Inhibits metabolism
- Reduces growth
- Destroys cells
- Causes DNA mutations
- Decline in crop productivity



OTHER EFFECTS

- UV rays accelerate the breakdown of paints and coatings.
- UV rays cause photo-chemical reactions which cause the production of harmful secondary pollutants like hydrogen peroxide (H_2O_2) which is known to have adverse effects on human health and plants.



OZONE DEPLETION

MEASURES TO PROTECT THE OZONE LAYER

VIENNA CONVENTION

- It is a multilateral environmental agreement. It was agreed upon at the Vienna Conference of 1985 and entered into force in 1988.
- It acts as a framework for the international efforts to protect the ozone layer.
- It does not include any legally binding steps for the reduction in the use of CFCs, the main substances responsible for ozone depletion. Such steps are laid out in the accompanying Montreal Protocol.



OZONE DEPLETION

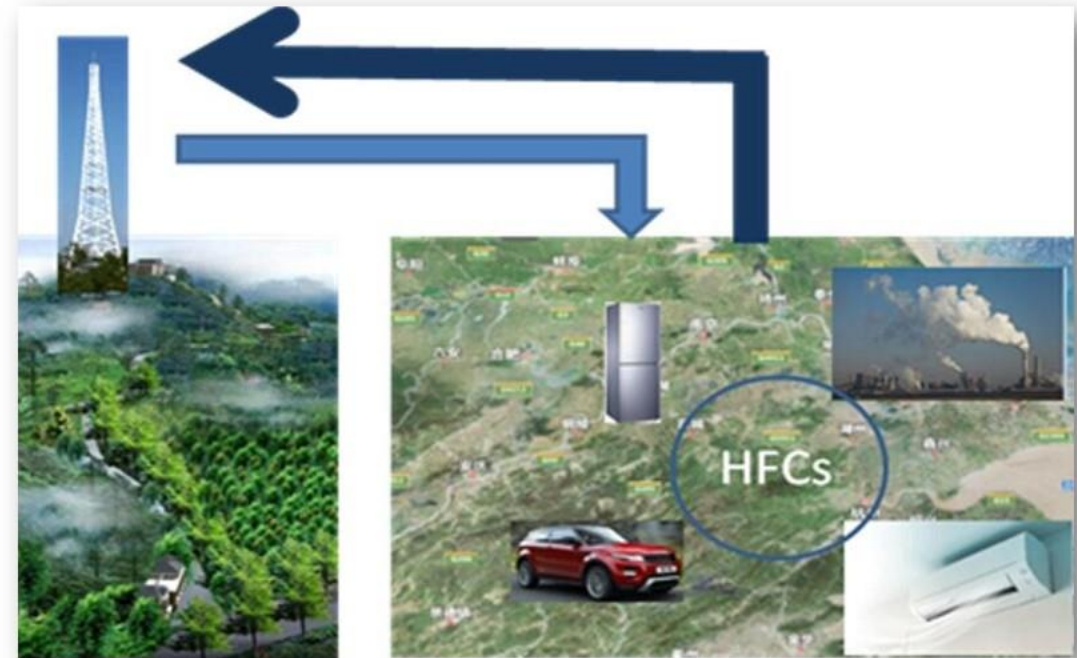
MEASURES TO PROTECT THE OZONE LAYER



MONTREAL PROTOCOL



- The Montreal Protocol on Substances that Deplete the Ozone Layer is a protocol to the Vienna Convention for the Protection of Ozone Layer.
- It is an international treaty designed to protect the Ozone Layer by phasing out the production of numerous substances that are responsible for ozone depletion.
- It was agreed in 1987 and came into force in 1989.
- Since then, it has undergone eight revisions. As a result of this protocol, the ozone hole in Antarctica is slowly recovering. Estimates indicate that the ozone layer will return to 1980 levels by 2050 or 2070.



OZONE DEPLETION

MEASURES TO PROTECT THE OZONE LAYER



MONTREAL PROTOCOL



- One of the outcomes of Montreal Protocol was substitution of CFCs with HCFCs. HydroChloroFluoroCarbons are also ozone depleting but less powerful compared to CFCs.
- Subsequently a decision was taken to replace HCFCs with HFCs (HydroFluroCarbons). These are produced mostly in developed countries.
- Hydro Fluoro Carbons replaced CFCs and HCFCs, they do not contain chlorine but are very powerful greenhouse gases.



OZONE DEPLETION

MEASURES TO PROTECT THE OZONE LAYER



KIGALI AMENDMENT

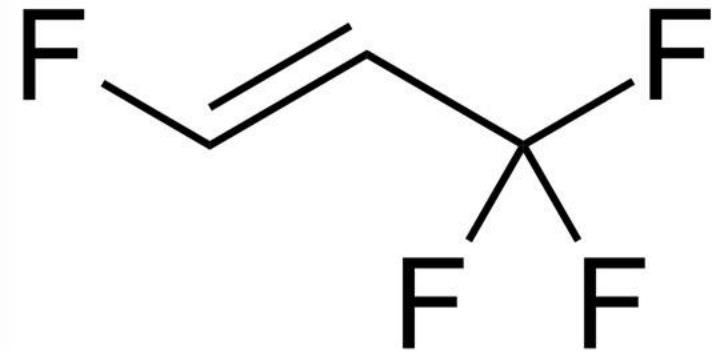


- In October 2016 at the 28th Meeting of the Parties to the Montreal Protocol in Kigali, the nations adopted the Kigali Amendment whereby they agreed to phase down the production and usage of HFCs.
- It is the 8th and the most recent amendment to Montreal Protocol. It is a legally binding agreement. It will come into effect from 1st January 2019.
- Under the Kigali Amendment to Montreal Protocol a decision was taken to replace HFCs with HFOs (Hydrofluoro Olefins). HFO has properties similar to HFCs.



MEASURES TO PROTECT THE OZONE LAYER

- Low toxicity.
- Low global warming potential.
- Zero ozone depletion potential.
- The Kigali Amendment has shown considerable flexibility in approach while setting phase-down targets for different countries.
- It has divided the signatory parties into three groups
 1. The first group consists of rich and developed economies like USA, UK and the EU countries who will start to phase down HFCs by 2019 and reduce it to 15% of 2012 levels by 2036.
 2. The Second Group consists of emerging economies like China, Brazil as well as some African countries who will start phase down by 2024 and reduce it to 20% of 2021 levels by 2045.
 3. The third group consists of developing countries and some of the hottest climatic countries like India, Pakistan, Iran, Saudi Arabia who will start phasing down HFCs by 2028 and reduce it to 15% of 2024-2026 levels till 2047.



OZONE DEPLETION

MEASURES TO PROTECT THE OZONE LAYER

ADVANTAGES OF HFOs

- Vienna Convention and Montreal Protocol together were universally ratified by all the member countries of the United Nations.
- They are considered to be the most successful international environmental agreements along with United Nations Framework Convention on Climate Change (UNFCCC) and United Nations Convention to Combat Desertification (UNCCD).



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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

98499 42299